

MOHD ANAS

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EDUCATION

B.Tech in Information Technology

IIIT Vadodara, Gujarat, India

August 2023 - May 2027

7.46 / 10.0 GPA

Coursework: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Database Management Systems

EXPERIENCE

Aarogya Kaya | Remote | Software Engineer Intern

July 2025 - January 2026

- **Architected production backend services** using CQRS, PostgreSQL, and Supabase RPCs, securing **15 RPC functions and database tables** with **JWT/RBAC, Row-Level Security, admin-only controls, and environment secrets**.
- Improved **production read performance for a service serving 1,000+ users daily**, reducing average latency from **420ms to 38ms** and maintaining **p95 below 250ms** through in-memory caching, Redis, and database access optimization.
- Engineered **fault-tolerant asynchronous workflows** with retries and idempotent processing, improving reliability of production operations under transient failures.
- Built **CI/CD quality gates** for backend and web services, automatically running **unit and Playwright E2E test suites**, linting, build validation, and deployment across pull requests and merges.

Open Source | Remote | Software Contributor

July 2025 - May 2026

- Migrated the **Next.js (Vercel) patch-next utility** from CommonJS to TypeScript ESM, saving **15–30 seconds** per local reinstall during development and testing.
- Contributed **22 merged pull requests and 12 issues** through **maintainer reviews, design discussions, and iterative feedback**, fixing bugs, implementing features, and improving developer tooling.

PROJECTS

A3.store – Distributed Key-Value Database | Python, AWS EKS, Kubernetes, Terraform, gRPC

[Live](#) | [Github](#)

- Built a **leaderless distributed key-value store** with **gRPC replication, quorum consistency, gossip-based failure detection, hinted handoff, and anti-entropy repair** across a **3-node, RF=2 cluster**.
- Designed **pluggable storage and peer-communication abstractions** using **Python ABCs, dependency injection, and SOLID principles**, enabling interchangeable implementations and isolated testing of distributed-system behavior.
- Deployed **3 stateful replicas on AWS EKS** with Kubernetes and Terraform, implementing **persistent storage, autoscaling, NetworkPolicies, PodDisruptionBudgets, readiness/liveness probes, and rolling updates** for resilient cluster operation.
- Designed a **Pytest unit and integration automation suite** covering **storage, quorum replication, gossip failure detection, hinted handoff, and anti-entropy recovery**, enforcing automated tests and **code-coverage gates in GitHub Actions**.

Second Brain – Agentic Career Path Simulator | Python, FastAPI, Redis Queue, LLMs

[Live](#) | [Github](#)

- Architected a **multi-stage AI reasoning backend** that transforms natural-language decisions into **persisted scenario graphs, tradeoffs, branching outcomes, and action plans** through pass-based LLM orchestration.
- Built asynchronous execution with **FastAPI and Redis/RQ background workers**, hardening LLM workflows with **multi-model fallback, structured-output parsing, malformed JSON recovery, and asynchronous HTTP client reuse**.

TECHNICAL SKILLS

Languages: Go, Python, C/C++, Java, JavaScript, SQL

Backend & Systems: REST APIs, gRPC, PostgreSQL, Redis, Distributed Systems, System Design

Cloud & DevOps: AWS, EKS, Kubernetes, Docker, Terraform, GitHub Actions, CI/CD

Testing & Observability: Pytest, Playwright, Unit Testing, Integration Testing, E2E Testing, Prometheus, Grafana

AI & LLM: LLM Orchestration, RAG, Agentic Workflows, Prompt Engineering, OpenRouter

ACHIEVEMENTS

- Ranked **365th** among **15,804** participants in CodeChef **START192D**.
- Ranked **3,026th** among **17,000+** participants in Codeforces Round 1110 (Div. 1 + Div. 2).

LEADERSHIP & ACTIVITIES

- Organized an intra-college hackathon and mentored **25+ teams** as a **Technical Committee Core Member**.
- Led recruitment, mentorship, and event initiatives for **100+ students** as **Joint Secretary of DOT Club**.